# Literature Survey on Gesture-Based Cryptographic Key Generation and AES-Based File Encryption

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## 1 Introduction

Recent research has explored biometric authentication, biometric-based cryptographic key generation, Key Derivation Functions (KDFs), and Advanced Encryption Standard (AES)-based file protection. These areas are directly related to the proposed GestCrypt system. The proposed system captures a predefined hand gesture through a camera, converts the recognized gesture into a reproducible cryptographic input, derives a 256-bit AES key using a Key Derivation Function, and uses the generated key for file encryption and decryption.

The objective of this literature survey is to compare recent research published after 2024 using parameters such as cryptographic key size, input/data type, plaintext size, key generation or KDF method, encryption algorithm, encryption and decryption time, computational complexity, resource consumption, security performance, and additional observations.

## 2 Literature Survey

The selected research papers are compared using cryptographic parameters, data characteristics, encryption techniques, computational complexity, performance, and security-related observations.

**Table 1: Comparison of Cryptographic and Data Characteristics**

| Author / Paper | Year | Input / Data Type | Key Size | Plaintext / Data Size | Key Generation / KDF / Encryption Technique |
|---|---|---|---|---|---|
| Szymoniak and Kubanek, *Biometry-based verification system with symmetric key generation method for Internet of Things environments* [1] | 2025 | Facial biometric images; camera-based verification | 256-bit entropy | Facial biometric dataset; 512×512 test-image resolution | Facial triangle coordinates combined with timestamps to generate a dynamic session key; symmetric encryption is used for secure communication |
| Yirga, Yirga and Addisu, *Cryptographic key generation using deep learning with biometric face and finger vein data* [2] | 2025 | Face and finger-vein biometric data | 128-bit | 425 biometric samples; face images resized to 160×160; 128 extracted features | FaceNet + VGG19 + PCA + Siamese Neural Network followed by fuzzy extraction and Goppa-code-based cryptographic key generation |
| Adinata, Fauzi and Khair, *Integration of Key Derivation Function (KDF) Development for Advanced Encryption Standard (AES) 256 Key Generator in Digital File Security* [3] | 2025 | PDF, DOCX, XLSX, PNG, MP3 and MP4 files | 256-bit derived key from 512-bit external key | Multiple file formats; exact test sizes not explicitly reported | 512-bit external key divided into two blocks, XOR processing and AES SubBytes transformation to generate a 256-bit key, followed by AES-256 file encryption |
| Pratama, Yuana and Ardinansyah, *Development and Evaluation of a Desktop Academic Data Encryption Application Using AES with Password-Based Key Management* [4] | 2026 | 10 authentic academic files | 128-, 192-, and 256-bit | 32,500 B–85,600,000 B | PBKDF2 followed by AES-128, AES-192, or AES-256 encryption |
| Sureshkumar et al., *A lattice-integrated AES framework for ultra-secure biometric protection on resource-constrained edge devices* [5] | 2026 | Fingerprint minutiae / biometric templates | 256-bit | 388×374 fingerprint images; 512-byte ISO biometric template | KDF-SHA-512 with 100,000 iterations, PIN input and LWE-based random salt, followed by AES-256-GCM |

**Table 2: Comparison of Encryption Performance and Security Characteristics**

| Author / Paper | Encryption Time | Decryption Time | Time Complexity | Resource Usage | Security / Performance / Additional Observations |
|---|---|---|---|---|---|
| Szymoniak and Kubanek [1] | Key generation: 0.8 s | Not reported | Not explicitly stated | Designed for resource-constrained IoT devices | 256-bit entropy; average randomness p-value = 0.98; estimated brute-force resistance $>10^{18}$ years. The system uses dynamic session-specific keys and liveness-aware biometric verification. |
| Yirga, Yirga and Addisu [2] | Not reported | Not reported | Polynomial complexity for Goppa-code decoding | Not explicitly reported | Sigma similarity = 92.66% for the 0.5 threshold vector test and sigma difference = 63.60%.<br>FAR < 1% and FRR < 3.4%. The study focuses mainly on security, accuracy and reproducibility rather than execution-time benchmarking. |
| Adinata, Fauzi and Khair [3] | Not reported | Not reported | Not explicitly stated | Desktop Python application | Successfully encrypts and decrypts multiple file formats. The main contribution is the KDF-based transformation of a 512-bit external input into a 256-bit AES key. |
| Pratama, Yuana and Ardinansyah [4] | AES-256: 2.45 s for an 85.6 MB file | Experimentally evaluated | O(n) for data processing | CPU: 0.1–0.2%<br>RAM: 53–54% | AES-128 provides higher performance, while AES-256 requires more processing.<br>The ciphertext introduces a constant 64-byte overhead consisting of salt, IV and padding.<br>Ciphertext entropy: 7.9988–7.9999<br>Avalanche effect: 49.85–50.19% |
| Sureshkumar et al. [5] | 0.020 s (final validation)<br>AES-GCM: 20.5 ms | 0.018 s (final validation)<br>AES-GCM: 18.3 ms | Feature extraction: O(n)<br>LWE: $O(n^2)$<br>MCC: O(N log N) | Minutiae extraction: 28.4 ms<br>ISO encoding: 4.1 ms<br>MCC matching: 15.7 ms | FVC2002 accuracy: 98.69%<br>SOCOFing accuracy: 98.54%<br>SOCOFing encryption/decryption: 0.022/0.020 s.<br>Total internal execution time is below 100 ms, supporting real-time resource-constrained deployment. |

# 3 Comparative Discussion

The five papers cover different components of secure cryptographic systems. Szymoniak and Kubanek [1] demonstrate that biometric information can be used as the basis for dynamic, session-specific symmetric-key generation. Their reported key-generation time of 0.8 seconds and 256-bit entropy demonstrate the feasibility of generating secure session keys from biometric information.

Yirga et al. [2] investigate reproducible cryptographic key generation from face and finger-vein biometric data using deep learning, feature fusion, fuzzy extraction, and Goppa-code cryptography. The reported low FAR and FRR values demonstrate strong biometric verification performance, although the paper primarily evaluates security and accuracy rather than encryption execution time.

Adinata et al. [3] focus directly on KDF development for AES-256 file security. Their approach transforms a 512-bit external input into a 256-bit derived key before AES-256 encryption, providing a direct reference for the KDF component of GestCrypt.

Pratama et al. [4] provide a detailed experimental comparison of AES-128, AES-192 and AES-256 using different academic file sizes. Their results demonstrate the trade-off between security and processing performance, with AES-256 requiring more computation while maintaining high ciphertext entropy and avalanche-effect values.

Sureshkumar et al. [5] combine biometric data, KDF-based key generation, lattice-derived randomness and AES-256-GCM. The system reports low encryption/decryption times together with high fingerprint detection accuracy, making it highly relevant to the proposed combination of biometric input and AES-256 encryption.

# 4 Research Gap

The reviewed studies demonstrate important advances in biometric authentication, biometric cryptographic key generation, KDF-based key derivation, and AES-based file protection. However, most existing systems concentrate on one or two of these components.

Biometric key-generation studies primarily use facial, fingerprint, or finger-vein characteristics. KDF-based studies focus on strengthening conventional passwords or external keys before AES encryption. Other works focus primarily on AES performance for different key sizes and file sizes.

The proposed GestCrypt system aims to combine these concepts into a single workflow:

**Hand Gesture Recognition → Reproducible Gesture Input → KDF → 256-bit AES Key → File Encryption**

Unlike the reviewed biometric systems, GestCrypt uses a predefined camera-based hand gesture as the input for cryptographic key regeneration. The system also aims to avoid directly storing the generated AES key and includes security evaluation through web application penetration testing.

# 5 Conclusion

The literature survey shows that biometric-based cryptographic key generation, KDF-based AES encryption, and performance evaluation of AES-based file protection are active research areas.

The selected papers demonstrate the feasibility of generating cryptographic keys from biometric inputs and the effectiveness of AES-256 and KDF techniques for digital-file protection.

However, there remains an opportunity to integrate camera-based predefined hand gesture recognition with reproducible key generation, KDF-based AES-256 file encryption, and web-application security evaluation in a single system. This motivates the proposed GestCrypt framework.

# References


[1] S. Szymoniak and M. Kubanek, "Biometry-based verification system with symmetric key generation method for internet of things environments," *Scientific Reports*, vol. 15, p. 5464, 2025.

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[4] M. W. R. Pratama, H. Yuana, and F. T. Ardinansyah, "Development and evaluation of a desktop academic data encryption application using aes with password-based key management," *Journal of Artificial Intelligence and Engineering Applications*, vol. 5, no. 3, pp. 4197–4210, 2026.

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